OFFICIAL ARCHITECTURE DOCUMENT

TrustLens Arc

Complete Plain-English Project Guide, Technical Architecture & Implementation Blueprint

Author: Sarfaraz **Category:** AI Security / Cyber-Forensics **Live URL:** trustlensarc.web.app

Date: 2026 Edition

1 What is TrustLens Arc? (In Plain English)

Imagine an intelligent security scanner that acts like an **antivirus for modern AI-driven fraud**. In today's digital landscape, cyber criminals have evolved beyond simple spam emails. They now deploy artificial intelligence to execute hyper-convincing scams:

1. CEO Voice Clones

Generative voice synthesizers (like ElevenLabs) clone executive voices from just 3-second samples to authorize urgent fake wire transfers over VoIP calls.

2. Job Offer Scams

Forged Google and Big Tech employment letters made with Canva that direct candidates to private Telegram handles and steal $2,450+ in fake equipment check schemes.

3. Quishing & Phishing Links

Weaponized QR codes that bypass conventional corporate email text filters, redirecting mobile users to typosquatted credential harvesting websites.

TrustLens Arc is an all-in-one web platform that detects, neutralizes, and detonates all three of these threats in real time — operating with sub-second latency directly inside the user's browser.

2 Live Production Links

🌐 Live Production Web Application:	`https://trustlensarc.web.app`
📊 Interactive Hackathon Pitch Slides:	`https://trustlensarc.web.app/presentation.html`
🐙 GitHub Repository (Source Code):	`github.com/itsayshaibrahim2006/Trustlens-Arc`

3 Technology Stack & Languages Used

🎨 User Interface & Frontend Core

- **JavaScript (ES6+)**
  : Main programming language for all UI logic, mathematical computations, and algorithms.
- **React 18**
  : Component-based architecture isolating the Hero, Scanner Bay, Audio suite, and Footer into maintainable units.
- **Vite**
  : Lightning-fast build tool and development server that hot-reloads and compiles production assets in under 14 seconds.
- **Tailwind CSS**
  : Utility-first CSS framework providing responsive layouts, dark cyber palettes, glassmorphism, and neon glows.

🎮 3D Graphics & Visual Telemetry

- **Three.js (WebGL)**
  : Renders the complete 3D Astro-Van vehicle with an astronaut pilot, metallic chassis, and spinning radar dish at 60 FPS without external plugins.
- **HTML5 Canvas 2D**
  : Powers the real-time plasma laser barrage, computing dynamic cubic bezier curves from the radar dish to inspection cards.

4 The Server & Backend Architecture (Why There is No Server!)

Zero-Server / 100% Client-Side Engine

Most websites send your sensitive files (resumes, voice clips, links) to a remote Python or Node.js server. That creates privacy violations and latency lag. **TrustLens Arc runs its entire forensic analysis directly on your computer inside your browser!**

- **Total Privacy (Zero PII Leakage)**: Uploaded resumes and audio never touch external third-party servers or databases.
- **Instant Speed (< 300ms)**: Calculations happen locally in milliseconds without upload waits.
- **Global Edge Hosting**: Hosted on **Firebase Hosting (Google Cloud)**, distributing compiled assets through Google's Worldwide Content Delivery Network (CDN) edge servers.

5 The 4 Main Features Explained in Detail

1️⃣ VoiceCheck 3D & CEO Deepfake Acoustic Spectrogram

The Problem:

Vishing fraudsters use ElevenLabs voice clones to authorize corporate payments.

How TrustLens Arc Solves It:

Uses the
Web Audio API

to decompose vocal harmonics in real time. Human vocal cords naturally vibrate with 0.8% - 2.5% fundamental pitch jitter (F0) and have respiratory diaphragm pauses. AI speech algorithms are mathematically too robotic (< 0.15% pitch jitter) and omit breathing pauses. If pitch is too flat, it immediately sounds the alarm.

2️⃣ Google Job Scam & CV / Resume Authenticator

The Problem:

Unsuspecting job hunters receive fake Google offer letters demanding money for home office equipment.

How TrustLens Arc Solves It:

Uses Mozilla's
PDF.js

to parse uploaded PDF/DOCX documents client-side. The engine audits the document's binary metadata (flagging Canva/Skia template generators), checks for advance-fee language ("*cashier check*", "*wire transfer*", "*equipment deposit*"), and intercepts unofficial communication channels (like unverified Telegram handles: *@google_careers_emea*). Legitimate candidates with verified GitHub repositories are awarded high authenticity scores.

3️⃣ Quishing Matrix & Sandbox Link Detonator

The Problem:

Scammers print QR codes on flyers or in emails to bypass security software.

How TrustLens Arc Solves It:

When a user uploads a QR code image,

jsQR

scans the pixel matrix on an off-screen canvas to extract the hidden URL. The link detonator runs the

Levenshtein Edit-Distance Algorithm

against 18+ major brands to catch typosquatting (e.g. *apple-support.tk* or *paypa1.com*), checks for Cyrillic IDN homoglyphs (lookalike characters), and flags dangerous top-level domains (.tk, .ru, .xyz).

4️⃣ Autonomous Scan Bay with 3D Astro-Van HUD

The Problem:

Security tools are often boring, static tables that users ignore.

How TrustLens Arc Solves It:

A complete Three.js 3D rover unit locks radar telemetry onto targets. Its plasma fire beams dynamically change color:

Crimson Red

when dangerous threats are confirmed, and

Luminescent Emerald Green

when credentials are authenticated.

6 How Your Friend Can Run This Project Locally

Follow these 5 simple terminal commands to run the entire project on any Mac, Windows, or Linux laptop:

> **Prerequisites:**
>
> Install **Node.js** (v18+) and **Git** if not already installed.

Step 1: Clone the GitHub Repository

```
git clone https://github.com/itsayshaibrahim2006/Trustlens-Arc.git
```

Step 2: Navigate into the Project Folder

```
cd Trustlens-Arc
```

Step 3: Install Required Dependencies

```
npm install
```

(Installs React, Three.js, Lucide icons, pdfjs-dist, and jsQR in ~15 seconds).

Step 4: Launch the Local Vite Server

```
npm run dev
```

Step 5: View the Application

```
Open your browser at: http://localhost:5173
```

7 Quick Reference Summary Matrix

Parameter	Project Specification
Project Name	**TrustLens Arc**
Mission Statement	Autonomous Multi-Modal Cyber-Forensic Defense Suite against 2026 AI Exploits
Core Threat Vectors	CEO Synthetic Voice Clones, Google Job Letter Forgery, Quishing / Malicious QR Links
Languages	JavaScript (ES6+), HTML5, CSS3
Frameworks & Tools	React 18, Vite Bundler, Tailwind CSS, Lucide React
3D Graphics Engine	Three.js WebGL (60 FPS hardware accelerated), HTML5 Canvas 2D
Forensics Libraries	PDF.js (Mozilla), jsQR (Computer Vision), Web Audio API (Acoustics)
Backend Philosophy	Zero-Server, 100% Client-Side Engine (Zero PII data exposure, <300ms speed)
Production Hosting	Firebase Hosting on Google Cloud Global Edge CDN
Live Production App	https://trustlensarc.web.app
Pitch Slide Deck	https://trustlensarc.web.app/presentation.html

TrustLens Arc © 2026 • Authored by Sarfaraz • Enterprise AI Cyber-Security